

2019 HKDSE Physics

Paper 1A — Multiple-choice Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Editable transcription of all question text, formulae, symbols and units. Only essential original figures are retained; repeated figure content is omitted.

Official answers are placed directly below each question.

Question 1

A block measuring $80\text{ }^{\circ}\text{C}$ is put into water at $40\text{ }^{\circ}\text{C}$. The final temperature of the mixture is $70\text{ }^{\circ}\text{C}$. Which deduction must be correct? Assume no heat loss to the surroundings.

- A. The energy gained by the water is greater than the energy lost by the block.
- B. The mass of the water is greater than the mass of the block.
- C. The specific heat capacity of water is smaller than that of the block's material.
- D. The heat capacity of the water is smaller than that of the block.

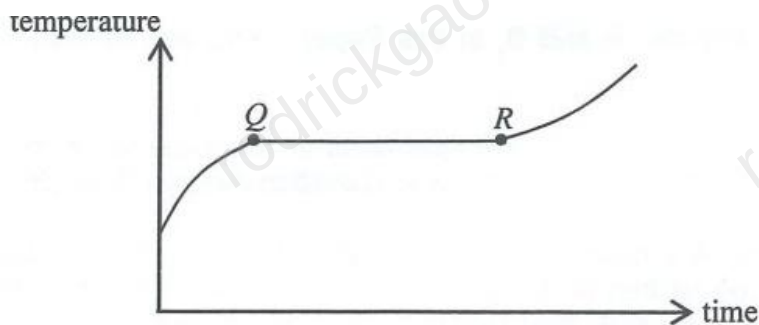
Official answer: D

With no heat loss, the block loses the same energy that the water gains. Hence $C_{\text{block}}(80-70) = C_{\text{water}}(70-40)$, so $C_{\text{water}} = C_{\text{block}}/3$ and the water has the smaller heat capacity. Answer: D.

Question 2

A substance undergoes fusion. Its temperature remains constant from O to R. Which deductions within this period are correct?

- (1) The substance does not absorb heat.
- (2) The mass ratio of the substance in solid and liquid states remains constant throughout.
- (3) The average potential energy of its molecules increases with time.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

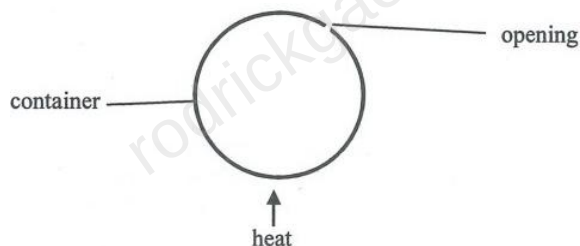
Official answer: B

During fusion the substance absorbs latent heat while its temperature stays constant. The solid-to-liquid mass ratio changes and molecular potential energy increases; only statement (3) is correct. Answer: B.

Question 3

The figure shows an inexpandable container with an opening. When the gas inside is heated slowly, which statements are correct?

- (1) The number of molecules decreases.
- (2) The average kinetic energy of the molecules increases.
- (3) The average separation between molecules remains unchanged.



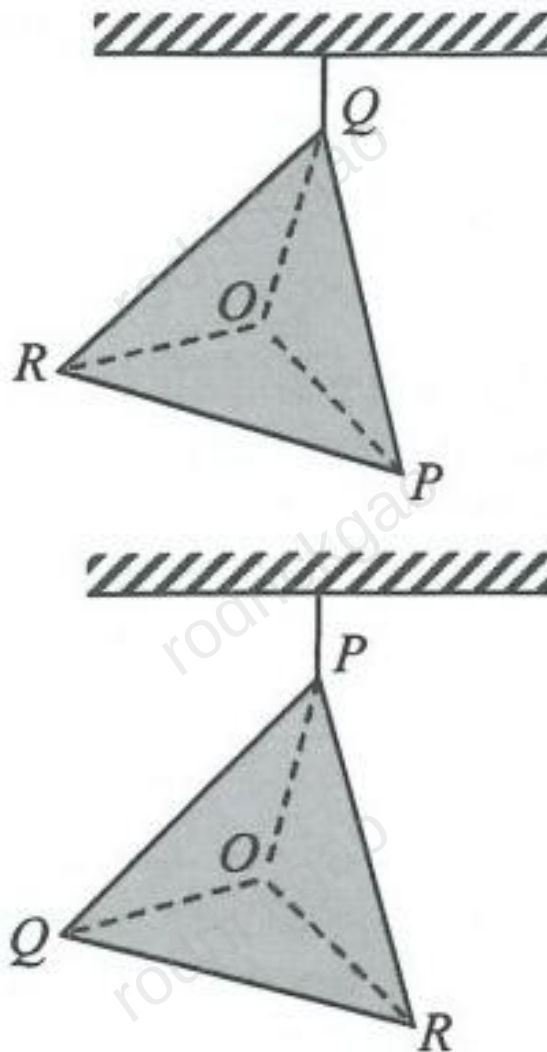
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Heating raises the molecular average kinetic energy. Because the container is open, gas expands and escapes, so the number of molecules decreases; the average separation need not remain unchanged. Thus (1) and (2) are correct. Answer: A.

Question 4

O is the centre of a non-uniform metal plate PQR in the form of an equilateral triangle. It is suspended from P and then from Q as shown. The centre of gravity is



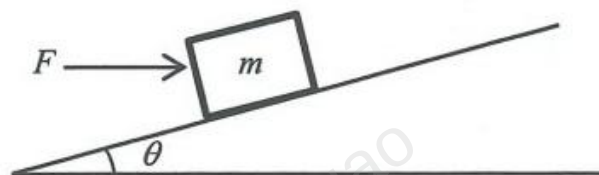
- A. at O.
- B. within region POQ.
- C. within region ROQ.
- D. within region POR.

Official answer: D

When suspended from P and Q, the vertical lines through the suspension points must both pass through the centre of gravity. Their intersection lies inside region POR. Answer: D.

Question 5

A horizontal force F keeps a block of mass m at rest on a smooth incline making an angle θ with the horizontal. Find F .



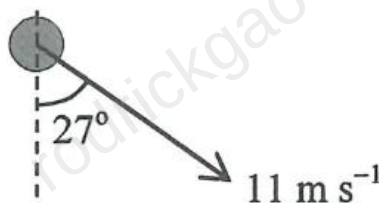
- A. $mg \sin \theta / \cos \theta$
- B. $mg \sin \theta \cos \theta$
- C. $mg \cos \theta / \sin \theta$
- D. $mg \sin \theta$

Official answer: A

Resolve forces along the smooth incline: the component of the horizontal force up the plane is $F \cos \theta$ and the component of weight down the plane is $mg \sin \theta$. Equilibrium gives $F = mg \tan \theta$. Answer: A.

Question 6

A small ball moves under gravity only. Its velocity at a certain instant is 11 m s^{-1} at 27° to the vertically downward direction, as shown. What was its speed 1 s earlier? Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)



- A. 19.1 m s^{-1}
- B. 9.8 m s^{-1}
- C. 5.0 m s^{-1}
- D. 0.2 m s^{-1}

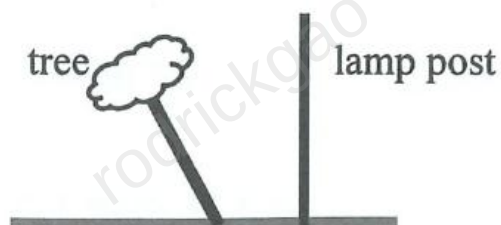
Official answer: C

The shown 11 m s^{-1} velocity has components $11 \sin 27^\circ \approx 5.0 \text{ m s}^{-1}$ horizontally and $11 \cos 27^\circ \approx 9.8 \text{ m s}^{-1}$ downward. One second earlier the vertical component was zero, so the speed was 5.0 m s^{-1} . Answer: C.

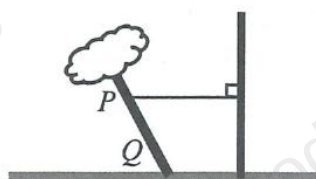
Question 7

A tree blown by strong wind leans to one side. A rope is wrapped around its trunk and tied to a fixed lamp post. In which arrangement would the rope have the highest chance of breaking?

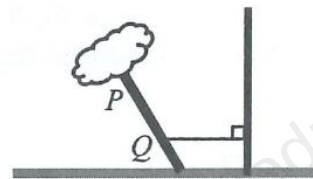
Options A–D are shown in the original figure.



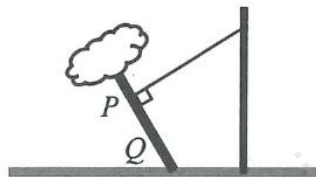
A.



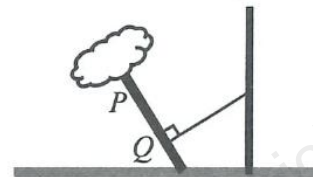
B.



C.



D.



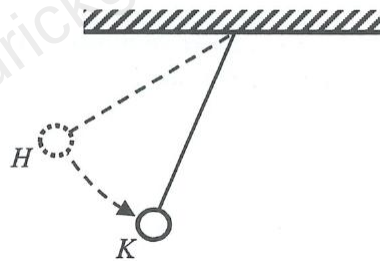
Official answer: B

For the same restoring moment on the leaning tree, rope tension is greatest when its perpendicular lever arm is smallest. Arrangement B gives the least effective moment arm and hence the highest breaking risk. Answer: B.

Question 8

A small sphere suspended by a light inextensible string is released from H. The string remains taut as the sphere swings down. Which free-body diagram shows all the forces acting on the sphere at K? Neglect air resistance.

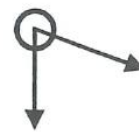
Options A–D are shown in the original figure.



A.



B.



C.



D.

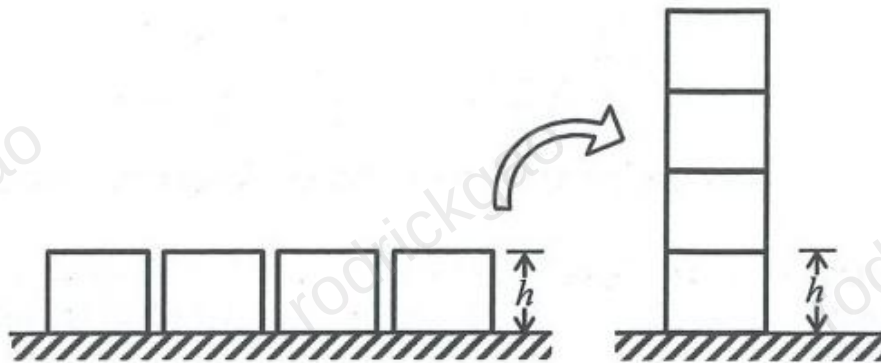


Official answer: D

At K the sphere is acted on only by its weight vertically downward and the string tension directed along the string toward H. The matching free-body diagram is D. Answer: D.

Question 9

Four identical uniform blocks, each of mass m and height h , are first placed on a horizontal table and then stacked. What is the minimum work done?



- A. $8mgh$
- B. $6mgh$
- C. $4mgh$
- D. $3mgh$

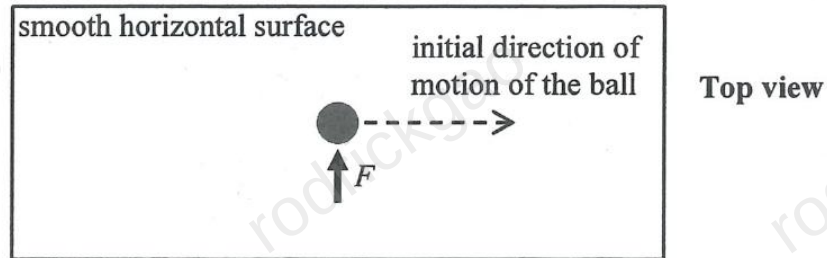
Official answer: B

Initially the four centres are each at $h/2$, so total GPE is $2mgh$. In the stack the centre heights are $h/2$, $3h/2$, $5h/2$ and $7h/2$, totaling $8mgh$. Minimum work = $6mgh$. Answer: B.

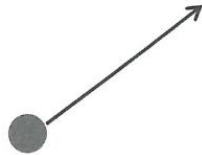
Question 10

A ball moves at constant speed along a straight line on a smooth horizontal surface. At a certain instant, a force F acts for a very short time as shown. Which subsequent path would the ball most closely follow?

Options A–D are shown in the original figure.



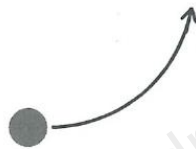
A.



B.



C.



D.



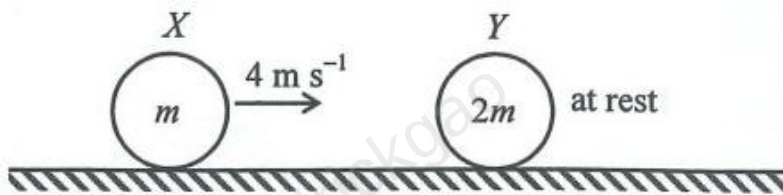
Official answer: A

The short force supplies an impulse in the direction of F , adding a transverse component to the original velocity. After the impulse the ball follows a straight line in the direction of the vector sum, as in A. Answer: A.

Question 11

A sphere X of mass m moving at 4 m s^{-1} collides head-on with a stationary sphere Y of mass $2m$. Which could be the speed of Y immediately after collision?

- (1) 1 m s^{-1} (2) 2 m s^{-1} (3) 3 m s^{-1}



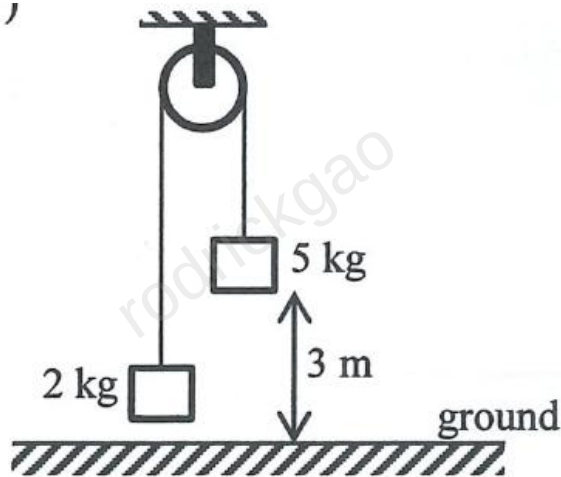
- A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

Official answer: B

Momentum gives $4m = mv_X + 2mv_Y$. If $v_Y = 1 \text{ m s}^{-1}$ then $v_X = 2 \text{ m s}^{-1}$, so X would still be catching Y; if $v_Y = 3 \text{ m s}^{-1}$ the final kinetic energy exceeds the initial value. Only $v_Y = 2 \text{ m s}^{-1}$ is possible. Answer: B.

Question 12

A 2 kg mass and a 5 kg mass are connected by a light inextensible string over a smooth pulley. The 5 kg mass is initially 3 m above the ground. Find its speed just before reaching the ground. ($g = 9.81 \text{ m s}^{-2}$)



- A. 5.0 m s^{-1}
- B. 6.0 m s^{-1}
- C. 6.5 m s^{-1}
- D. 7.7 m s^{-1}

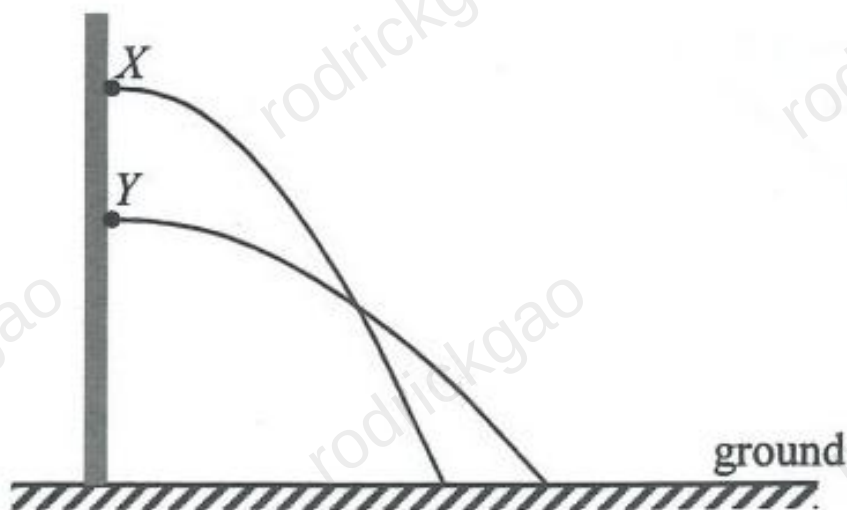
Official answer: A

The Atwood acceleration is $a = (5-2)g/(5+2) = 3g/7$. Starting from rest through 3 m, $v = \sqrt{2as} = \sqrt{(18g/7)} \approx 5.0 \text{ m s}^{-1}$. Answer: A.

Question 13

Particles X and Y are projected horizontally from a vertical wall and follow the paths shown. Which statements are correct?

- (1) The time of flight of Y is longer than that of X.
- (2) The projection speed of Y is greater than that of X.
- (3) X and Y can have the same speed on reaching the ground.



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

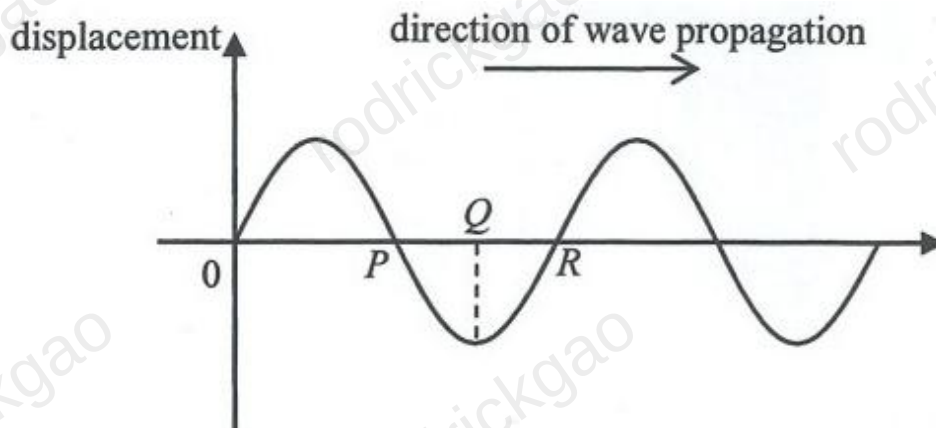
Official answer: C

Both fall through the same vertical height, so their times of flight are equal; statement (1) is false. Y travels farther horizontally, so its projection speed is greater, and suitable horizontal speeds can make the two landing-speed magnitudes equal. Thus (2) and (3) are correct. Answer: C.

Question 14

A longitudinal wave travels to the right. The graph shows displacement against distance at an instant. Which statements are correct?

- (1) P is at a compression.
- (2) Q is momentarily at rest.
- (3) R is moving downwards.



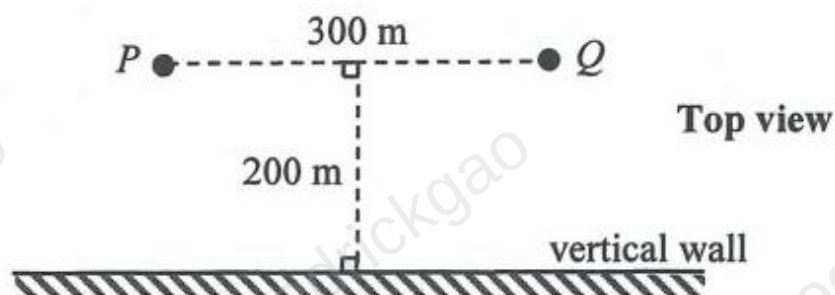
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

For a longitudinal displacement–distance graph, compression occurs where the displacement gradient is negative, so P is a compression centre. Q is at an extreme displacement and is instantaneously at rest; the motion statement at R is false. Thus (1) and (2) are correct. Answer: C.

Question 15

Boys P and Q are 300 m apart and each is 200 m from a reflecting wall. P shouts once. Which description of the two sounds heard by Q is correct? Speed of sound = 340 m s^{-1} .



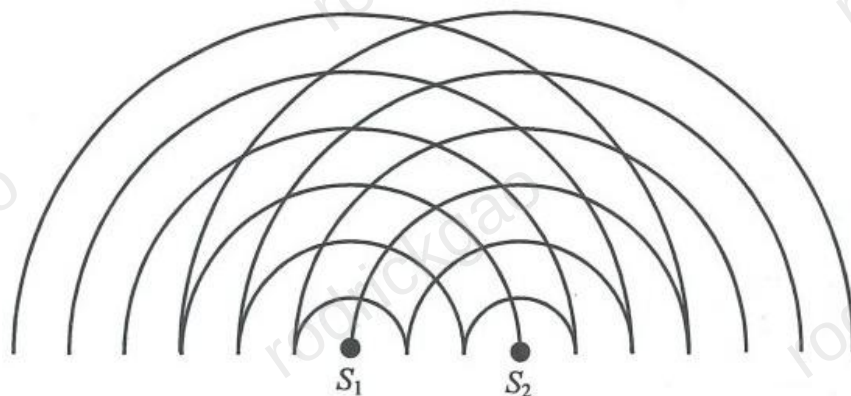
- A. The first is louder; the second arrives 0.59 s later.
- B. The first is louder; the second arrives 0.29 s later.
- C. The second is louder; the first arrives 0.59 s earlier.
- D. The second is louder; the first arrives 0.29 s earlier.

Official answer: A

The direct path PQ is 300 m. Reflecting P in the wall gives the echo path $\sqrt{(300^2 + 400^2)} = 500 \text{ m}$, so the delay is $(500 - 300)/340 = 0.588 \text{ s} \approx 0.59 \text{ s}$. The shorter direct sound is louder. Answer: A.

Question 16

Two in-phase dippers S_1 and S_2 produce circular water waves. How many nodal lines are formed?



- A. 3
- B. 4
- C. 6**
- D. 7

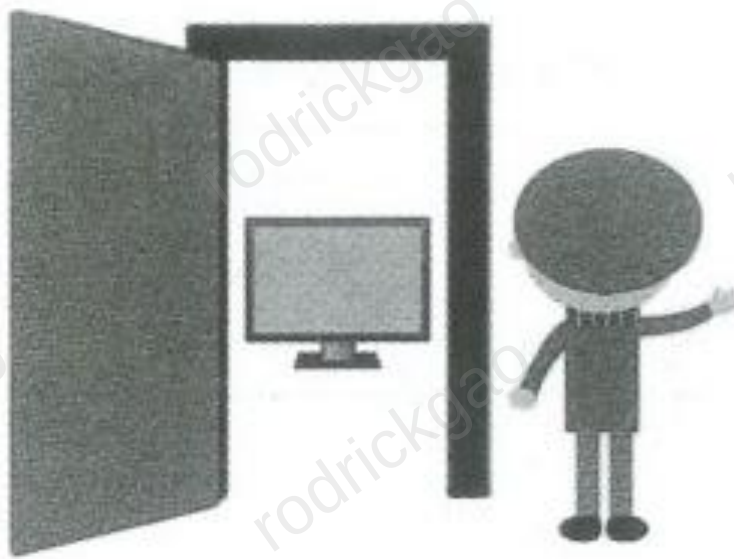
Official answer: C

Nodal lines are the loci where waves from the in-phase dippers arrive with path difference $(n+1/2)\lambda$. Counting the destructive-interference branches in the crest diagram gives three nodal lines. Answer: C.

Question 17

A television in an adjacent room can be heard but not seen. Which explanations are correct?

- (1) Sound undergoes noticeable diffraction at the doorway whereas light does not.
- (2) Sound is mechanical whereas light is electromagnetic.
- (3) The wavelength of sound is much longer than that of light.



- A. (1) only
- B. (3) only**
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

Both sound and light diffract, and their mechanical/electromagnetic nature does not explain the observation.

Sound has a wavelength comparable with the doorway and diffracts appreciably, whereas visible light has a tiny wavelength. Only (3) is correct. Answer: B.

Question 18

A transverse wave travels on a stretched string. How does the wave speed change when (i) the tension is increased and (ii) the string is replaced by a more massive string of the same length under the same tension?

Option	Tension increased	More massive string
A	increases	decreases
B	increases	increases
C	decreases	decreases
D	decreases	increases

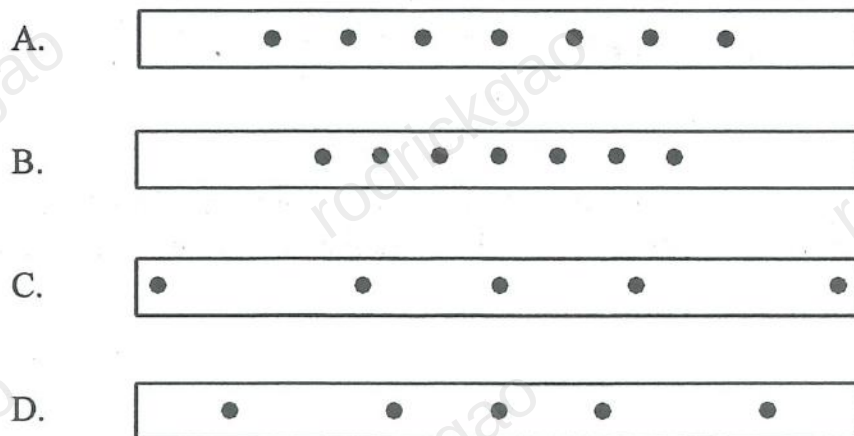
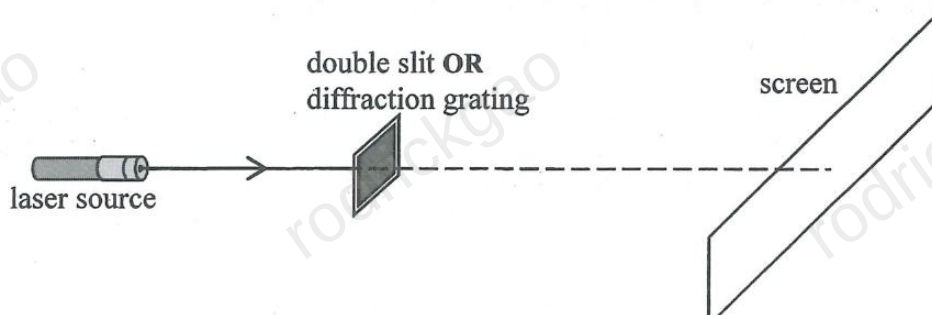
Official answer: A

Wave speed on a stretched string is $v = \sqrt{T/\mu}$. Increasing tension increases v , while replacing the string by a more massive one of the same length increases μ and decreases v . Answer: A.

Question 19

A double slit and a diffraction grating are used in turn with red and green laser light. Which pattern belongs to green light incident on the diffraction grating?

Options A–D are shown in the original figure.

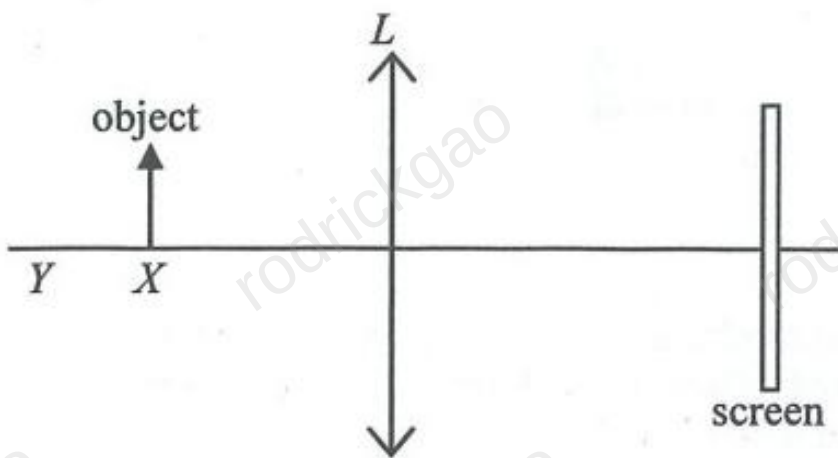


Official answer: D

A diffraction grating produces narrow, sharply defined principal maxima. Green light has a shorter wavelength than red light, so its maxima have the smaller angular spacing; pattern D has both features. Answer: D.

Question 20

An object is first placed at X in front of a convex lens and a sharp image is formed on a screen. It is then shifted to Y as shown. Which change can restore a sharp image on the screen?



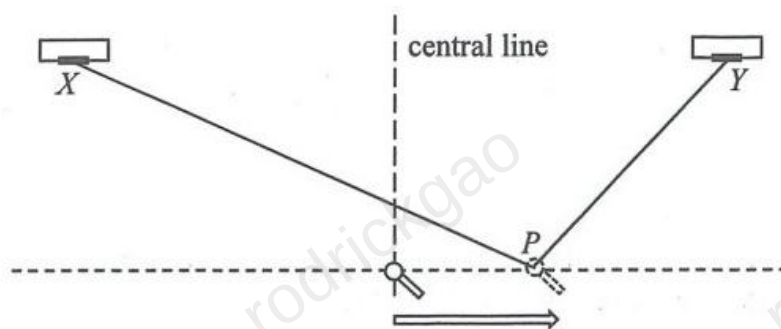
- A. Replace the lens by a convex lens of longer focal length.
- B. Use a lens of the same shape made from material of higher refractive index.
- C. Replace the lens by a concave lens.
- D. Move the screen to the right.

Official answer: A

With the screen fixed, shifting the object from X to the more distant point Y increases u . From $1/f = 1/u + 1/v$ with v fixed, f must increase, so a convex lens of longer focal length can refocus the image. Answer: A.

Question 21

Two coherent speakers X and Y emit 500 Hz sound. A microphone is moved along the line shown. P is the next position of maximum intensity. Find $PX - PY$. Speed of sound = 340 m s^{-1} .



- A. 0.17 m
- B. 0.34 m
- C. 0.51 m
- D. 0.68 m

Official answer: D

Adjacent maxima from the two coherent loudspeakers differ in path by one wavelength. $\lambda = v/f = 340/500 = 0.68 \text{ m}$, so $PX - PY = 0.68 \text{ m}$. Answer: D.

Question 22

Which statements about infrared radiation are correct?

- (1) It bends towards the normal when travelling from air into water.
- (2) It travels faster in water than in air.
- (3) It can be used for satellite communication.

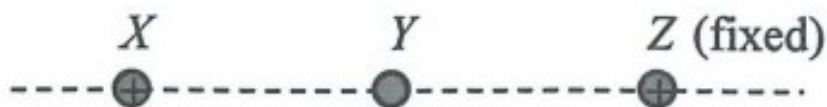
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

Infra-red is an electromagnetic wave: it slows in water and bends toward the normal when entering from air. It does not travel faster in water, and satellite communication normally uses microwaves. Only (1) is correct. Answer: A.

Question 23

Charges X and Z are identical positive charges. Y is at the midpoint. X and Y are in equilibrium while Z is fixed. If Y is pushed slightly towards Z and released, it will



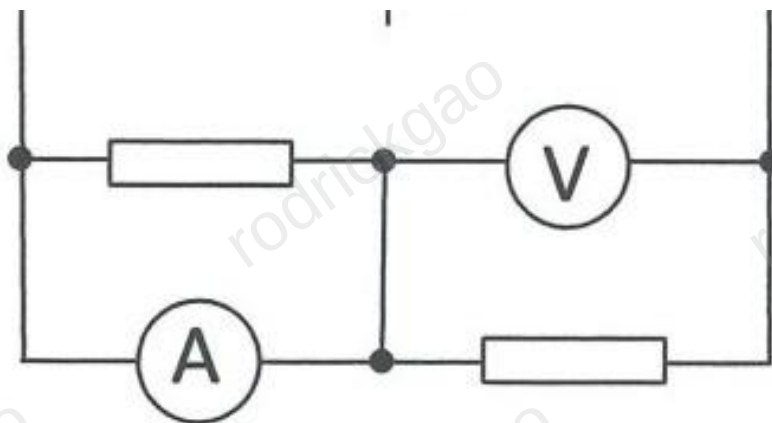
- A. move to the left.
- B. move to the right.
- C. remain at the new position.
- D. not be predictable because the sign of Y is unknown.

Official answer: A

Because X and Z are positive, equilibrium requires Y to be negative. Moving Y toward Z increases its distance from X, so its attraction on X weakens while Z's leftward repulsion is unchanged; X moves left. Answer: A.

Question 24

In the circuit shown, the cell has e.m.f. 6 V and the three resistors are identical. The ammeter and voltmeter are ideal. What is the voltmeter reading?



- A. 6 V
- B. 4 V
- C. 3 V
- D. 2 V

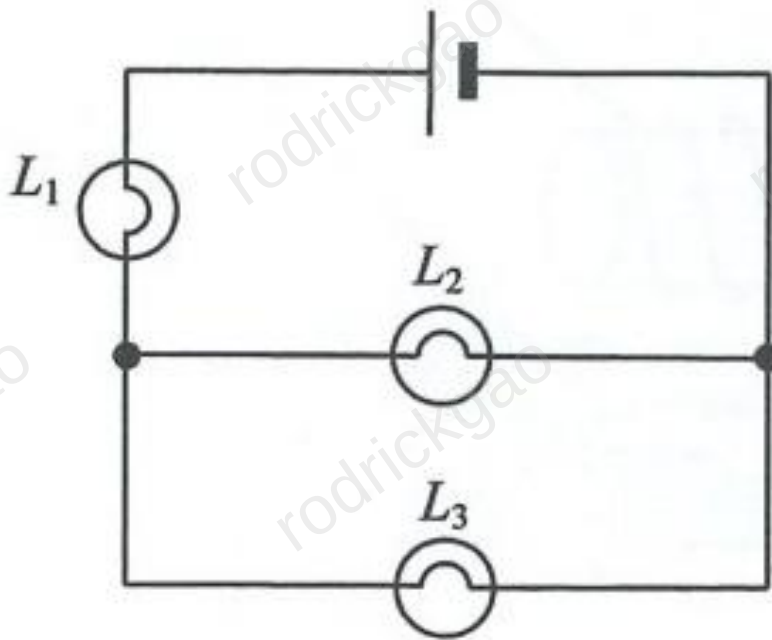
Official answer: C

The ideal ammeter acts as a wire and the ideal voltmeter draws no current. The conducting circuit therefore leaves two identical resistors sharing the 6 V supply equally, so the voltmeter reads 3 V. Answer: C.

Question 25

L_1 , L_2 and L_3 are light bulbs and the cell has negligible internal resistance. Which changes make L_3 brighter?

- (1) L_1 becomes a short circuit.
- (2) L_2 becomes a short circuit.
- (3) L_2 becomes an open circuit.



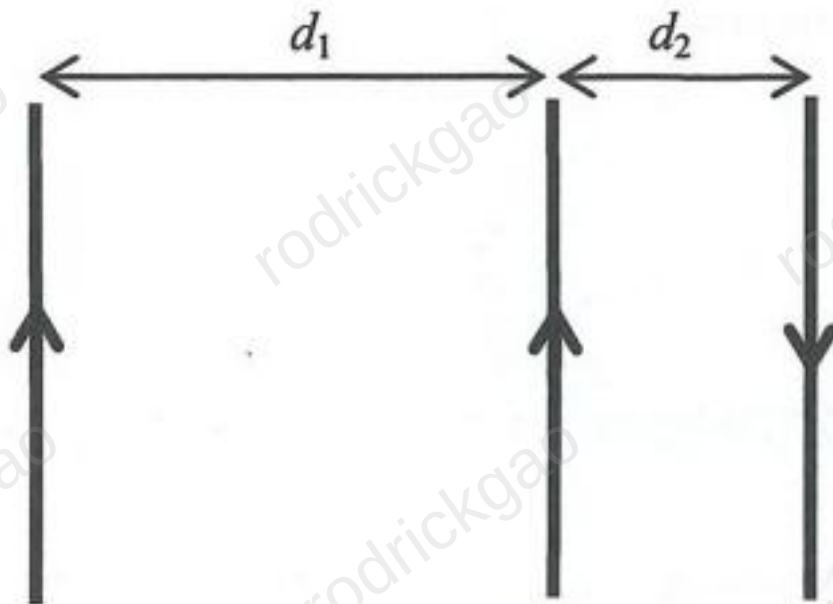
- A. (1) and (2) only
- B. (1) and (3) only**
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

L_1 is in series with the parallel pair L_2 – L_3 . Shorting L_1 increases the voltage across L_3 ; opening L_2 raises the parallel-section resistance and its voltage. Shorting L_2 instead shorts L_3 . Thus (1) and (3) are correct. Answer: B.

Question 26

Three parallel wires P, Q and R have separations d_1 and d_2 , with $d_1 > d_2$, and carry equal currents in the directions shown. The magnetic force per unit length on Q due to P is F . What are the direction and magnitude of the resultant force per unit length on Q?



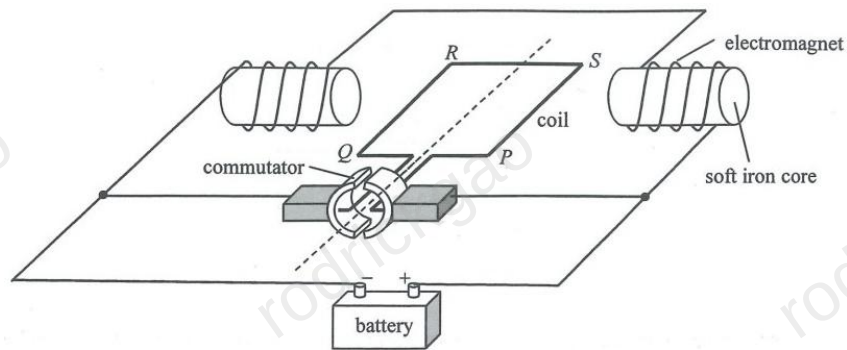
- A. to the right / greater than $2F$
- B. to the right / smaller than F
- C. to the left / greater than $2F$
- D. to the left / smaller than F

Official answer: C

P and Q carry currents in the same direction, so P attracts Q leftward with force F . Q and R carry opposite currents, so R also pushes Q leftward; because $d_2 < d_1$ this force exceeds F . The resultant is leftward and greater than $2F$. Answer: C.

Question 27

A motor coil PQRS is placed between electromagnets and connected to a d.c. supply. If the battery is replaced by a 50 Hz sinusoidal a.c. supply, the coil will



- A. remain at rest.
- B. oscillate at 50 Hz.
- C. rotate until vertical and stop.
- D. rotate continuously.

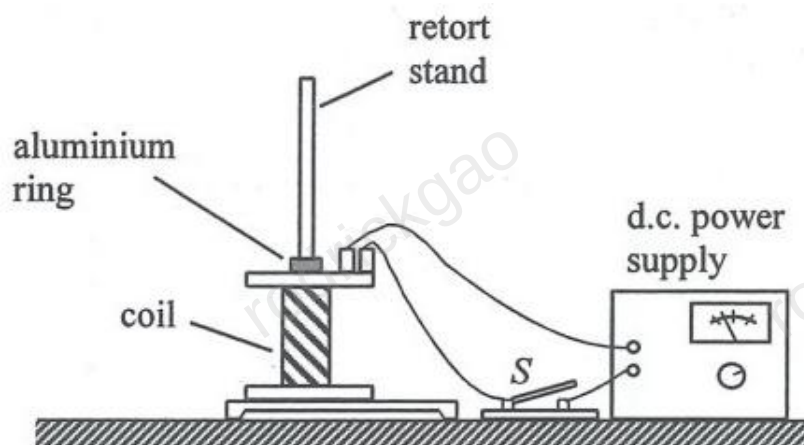
Official answer: D

With an a.c. supply, the current in the armature coil and the electromagnets reverses simultaneously every half-cycle. Both magnetic fields reverse, so their torque keeps the same direction and the coil rotates continuously.

Answer: D.

Question 28

An aluminium ring placed over a coil carrying d.c. jumps momentarily when the switch is closed. Which modification can make the ring float continuously?



- A. Use a lighter aluminium ring.
- B. Use a ring of smaller resistivity.
- C. Double the number of turns of the coil.
- D. Use an a.c. supply.

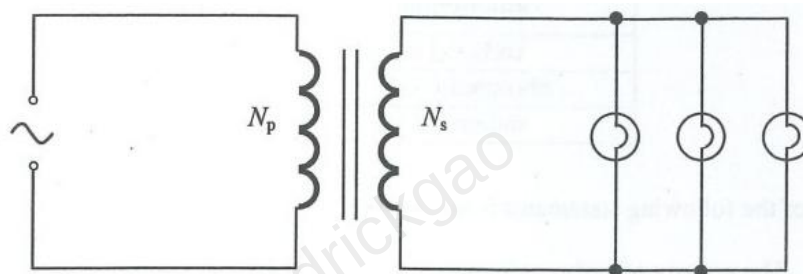
Official answer: D

A d.c. switch produces only a momentary flux change, so the induced current and lift decay. An a.c. supply continually changes the flux and sustains the opposing induced force, allowing the ring to rise and float. Answer:

D.

Question 29

A transformer is connected to 240 V a.c. Three 12 V, 24 W bulbs are connected in parallel across the secondary. What should the primary-to-secondary turns ratio $N_p:N_s$ be?



- A. 40:1
- B. 30:1
- C. 20:1
- D. 10:1

Official answer: C

The three bulbs are in parallel and each requires 12 V, so the transformer must step 240 V down to 12 V. For an ideal transformer, $N_p/N_s = V_p/V_s = 240/12 = 20$. Answer: C.

Question 30

The power delivered to an a.c. heater can be increased by which changes?

- (1) Increase the resistance of the heater.
- (2) Increase the frequency of the supply.
- (3) Increase the r.m.s. voltage of the supply.

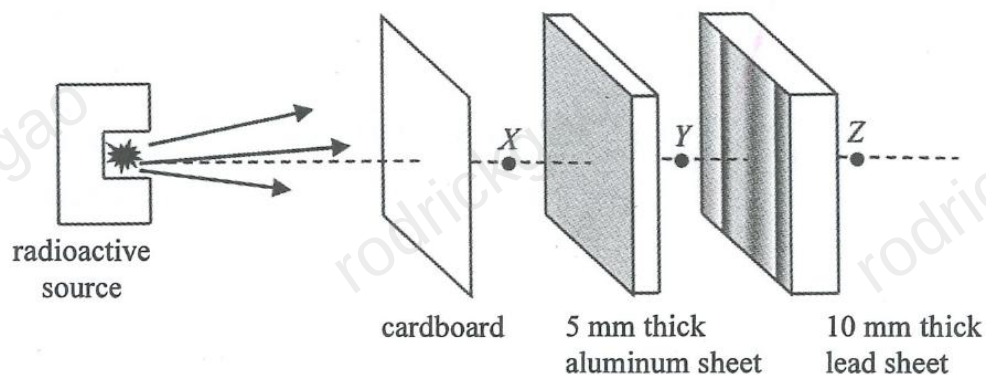
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

For a resistive heater, $P = V_{\text{rms}}^2/R$. Increasing R lowers the power and changing frequency does not affect it ideally, while increasing V_{rms} raises it. Only statement (3) is correct. Answer: B.

Question 31

A source emits α , β and γ radiation. Cardboard, aluminium and lead absorbers are arranged as shown. Which statement is correct?



- A. No radiation from the source is detected at Z.
- B. Both β and γ radiation are detected at Y.
- C. Only α radiation is detected at X but not at Y or Z.
- D. Only β radiation is detected at X but not at Y or Z.

Official answer: D

Cardboard stops α radiation before X; β can pass the cardboard but is stopped by the 5 mm aluminium sheet, so it is detected at X only. γ can penetrate farther, including toward Z. Answer: D.

Question 32

The table gives half-lives of three radionuclides. Which statements are correct?

- (1) The activity of a carbon-11 sample must be the highest.
- (2) The decay constant of phosphorus-32 is larger than that of carbon-11.
- (3) If the initial activity of sodium-22 is 1520 Bq, its activity is lower than 380 Bq after 6 years.

Radionuclide	Half-life
carbon-11	20.3 min
phosphorus-32	14.3 days
sodium-22	2.60 years

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

Activity depends on both decay constant and number of nuclei, so (1) is not guaranteed; phosphorus-32 has a smaller decay constant than carbon-11. For sodium-22, $A = 1520(1/2)^{6/2.60} \approx 307 \text{ Bq} < 380 \text{ Bq}$. Only (3) is correct. Answer: B.

Question 33

Given neutron mass = $16749 \times 10^{-31} \text{ kg}$, proton mass = $16726 \times 10^{-31} \text{ kg}$ and electron mass = $9 \times 10^{-31} \text{ kg}$, estimate the energy released when a neutron decays into a proton and a β particle.

- A. 1.8 MeV
- B. 1.3 MeV
- C. 0.79 MeV
- D. 0.51 MeV

Official answer: C

For $n \rightarrow p + \beta^-$, $\Delta m = m_n - m_p - m_e \approx 1.39 \times 10^{-30} \text{ kg}$. Hence $\Delta E = \Delta mc^2 \approx 1.25 \times 10^{-13} \text{ J} = 0.78 \text{ MeV} \approx 0.79 \text{ MeV}$. Answer: C.